TSITA MOGAU SETSHABA PHIAS

Database Developer Intern

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my-portfolio Linkedin Github

SUMMARY

BSc (Computer Science & Statistics) graduate with demonstrated expertise in database development, data management, and analytical systems. Successfully delivered solutions that reduced data inconsistencies by 83% and improved system response times by 61%. Proficient in SQL, Python, Excel, and data visualization tools with strong analytical and problem-solving abilities. Experienced in database design, data integrity validation, and building dashboards for reporting. Eager to apply technical skills in a data-driven environment while contributing to SANAC's mission of enhancing data integrity, reporting capability, and operational efficiency. Committed to learning and growing within a dynamic team focused on public health impact.

EXPERIENCE

University of Limpopo

4th Annual Limpopo Varsity Hackathon

10/2025 – 10/2025 | Limpopo, Mankweng

- Developed Smart Green House prototype using React and Python for remote crop monitoring
- Trained text-to-text model for automated environmental condition management.
- Collaborated with team of 4 developers over 3-day intensive hackathon.
- Received Certificate of Appreciation for contribution to innovative IoT solution

MLAB

React Developer

04/2024 – 11/2024 | Limpopo, Polokwane

- Front-End Skills: Built dynamic websites and apps with React.js, React Native, JavaScript, TypeScript, HTML, and CSS.
- Backend Proficiency: Used Firebase for real-time data, authentication, and cloud storage.
- User-Focused: Delivered scalable, user-friendly solutions tailored to specific needs.
- Version Control: Used Git/GitHub Desktop to streamline development and collaboration.

PROJECTS

End-to-end model for xitsonga language

Machine Learning Project

02/2025 – 11/2025

•Developed and trained a Whisper model for Xitsonga speech recognition with cleaned data.
•Used Python and the Hugging Face Transformers library
•Leveraged the NCHLT dataset for training data
•Achieved a Word Error Rate (WER) of 10%, improving speech-to-text accuracy for this low-resource language

Database Management

02/2025 – 05/2025

- Designed a centralized database system for managing projects, employees, clients, inventory, and IT support tickets.
- Created the ER diagram, normalized database schema, and data dictionary following DDLC steps.
- Developed sample SQL queries to support reporting and key business operations.

Cloud OS Process Synchronization.

02/2025 – 02/2025

- Built a hybrid synchronization system reducing data inconsistencies by 83% and improving response times by 61%, managing 50 concurrent threads with 10,000+ operations.

- Scaled production workloads to 3x peak traffic using Redis-based distributed locks, priority queuing, and circuit breaker patterns.

SKILLS

Front-End
- HTML
- CSS
- REACT
- JAVASCRIPT

Database
- PostgreSQL
- SQL
- Firebase
- MongoDB

Back-end
- NODE JS
- EXPRESS
- PYTHON

Mobile Development
- React Native

Tools
- Git
- Github desktop
- Hugging Face
- Trello
- analytical skills
- Power BI
- Excel
- Information Systems
- system documentation
- technical assistance

EDUCATION

University of Limpopo
Bachelor of Science in Computer Science
02/2022 – 12/2025 | Polokwane, Mankweng

KGOPUDI SECONDARY SCHOOL
National Senior Certificate
2021 | Bochum
Mathematics: 81%

ACHIEVEMENTS/CERTIFICATES

Network Micro-Certification
Completed Network Layer Protocols Micro-Certification with comprehensive understanding.

Hackathon competition
Earned participation certificate in 3-day Hackathon competition with team leadership.

Digital Literacy Made Easy - NEMISA
Certificate of Attendance for Introduction to Digital Literacy, Microsoft Office and Emerging Technologies

LANGUAGES

Sepedi

English

VOLUNTEERING

Computer Science Lab Assistance Volunteer
06/2023 – 08/2023

- Provided technical support to students in Java (Eclipse IDE), C++, and SQL while maintaining lab systems and ensuring smooth software operations.
- Collaborated with faculty to prepare lab materials and fostered a conducive environment for hands-on problem-solving and skill development.

REFERENCES

Mr Nhlanhla Chauke, *Supervisor,* MLab
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Mr Malatji, *Machine Learning Supervisor,*
University of Limpopo
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Dr Baloyi, *Hackathon Organiser,*
University of Limpopo
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